Connected Component Analysis for Multi-Vehicle Detection in Shipment Tracking Systems

B.Tech Project Presentation

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Problem & Motivation

GPS-based shipment tracking generates **millions of records daily**. A critical data quality failure: a single shipment tracked by **multiple vehicles simultaneously**.

Root causes: device misconfiguration, data integration errors, duplicate transmissions, system migration issues.

Observable symptoms:

- **Impossible teleportation** — Shanghai to Shenzhen (1,200 km) in 5 min
- **Oscillating locations** — jumping between distant areas
- **Concurrent distant positions**

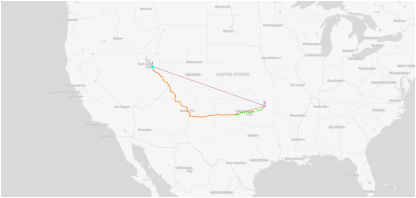
Business Impact

- Customer confusion
- Incorrect ETAs
- Billing disputes
- Operational blindness
- SLA violations

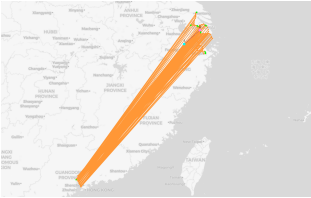
Challenge

Must distinguish errors from **legitimate handovers** (e.g. truck → ship at port).

The Problem in Practice

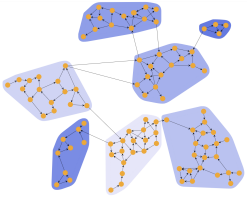


A typical GPS-based shipment tracking interface



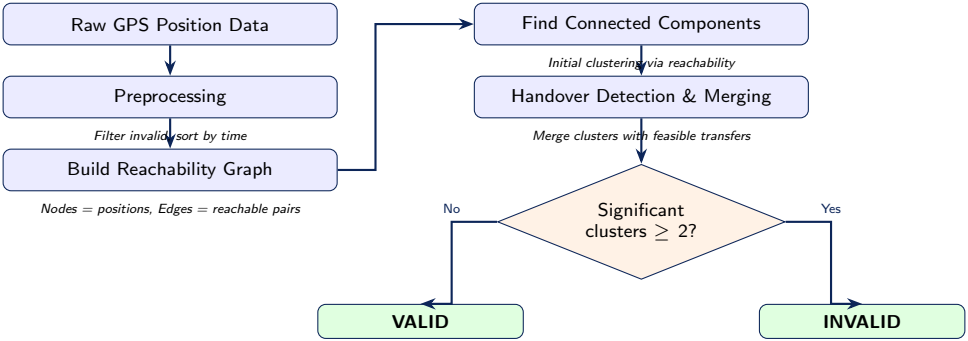
Before: Conflicting GPS trails from two devices

After: Trails separated, anomaly identified



Connected components color-coded by cluster

Proposed Framework: System Overview



Graph Theory Background

Fundamental Definitions

- **Graph** $G = (V, E)$: vertices + edges
- **Directed graph**: edge $(u, v) \neq (v, u)$
- **DAG**: directed, no cycles
- **Connected component**: maximal subgraph where every pair is connected

Haversine Distance

$$d = 2R \cdot \arcsin\sqrt{\sin^2\frac{\Delta\phi}{2} + \cos\phi_i \cos\phi_j \sin^2\frac{\Delta\lambda}{2}}$$

$R = 6{,}371$ km. Error $< 0.3\%$ vs. ellipsoidal.

Notation

$p_i = (\text{lat}, \text{lon}, t)$	Position
$P = \{p_1, \ldots, p_n\}$	Sorted sequence
$V_{\max} = 69.4$ m/s	Max speed
$\tau = 1800$ s	Jitter window
$T_H = 28800$ s	Handover window
$\alpha = 0.10$	Min cluster %
$s_{\min} = 5$	Min cluster size

Reachability

p_2 reachable from p_1 iff:

$$d(p_1, p_2) \leq V_{\max} \cdot (t_2 - t_1)$$

Graph Formulation

Position Reachability Graph $G = (V, E)$

- Each GPS position $p_i = (\text{lat}, \text{lon}, t)$ is a **vertex**
- Edge (p_i, p_j) exists iff movement is physically feasible:

$$\text{reachable}(p_i, p_j) \iff \underbrace{d(p_i, p_j)}_{\text{Haversine}} \leq V_{\max} \cdot (t_j - t_i), \qquad V_{\max} = 250 \text{ km/h}$$

Jitter Tolerance

GPS noise handled by checking **multiple recent** positions within $\tau = 30$ min window:

$$\exists\, p_i \in W(p_j, C, \tau) : \text{reachable}(p_i, p_j)$$

Handover Detection

Legitimate transfers (truck $\rightarrow$ ship) detected via boundary positions of adjacent clusters:

$$|t_i - t_j| \leq 8\text{h} \;\wedge\; \text{reachable}(p_i, p_j)$$

Key Theorems & Properties

Theorem 1: Single Component $\Rightarrow$ Single Vehicle

If the Position Reachability Graph has exactly one weakly connected component, all positions are consistent with single-vehicle tracking.

Theorem 2: Multiple Large Components $\Rightarrow$ Multiple Vehicles

If G has ≥ 2 components each containing $\geq 10\%$ of total positions, multiple vehicles are tracking the shipment.

Theorem 3: Handover Preserves Validity

If two components can be merged via a valid handover, the merged component represents valid single-vehicle tracking (with transfer).

Graph properties: always a DAG (edges follow time), reachability is **non-transitive**, significance filters noise: $\text{significant}(C) \iff |C| \geq \max(5,\ 0.10 \cdot n)$

Algorithm: SingleVehicleCheck

Phase 1: Preprocessing
Filter nulls/NaN, validate ranges, sort

↓

Phase 2: Clustering
Online reachability clustering + handover merging

↓

Phase 3: Decision
Count significant clusters → result

Clustering (Phase 2)

For each position chronologically:

1. Check recent positions in existing clusters (jitter window τ, up to 3)
2. **Reachable?** → assign to cluster
3. **Not?** → new cluster

Then merge clusters via handover predicate.

Decision (Phase 3)

0 significant → UNCERTAIN
1 significant → TRUE
≥ 2 significant → FALSE

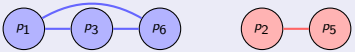
Worked Example: Shanghai & Shenzhen

Input (6 GPS positions)

p_1:	Shanghai	31.23°N	08:00
p_2:	Shenzhen	22.55°N	08:15
p_3:	Shanghai	31.24°N	09:00
p_4:	NaN		09:30
p_5:	Shenzhen	22.56°N	10:00
p_6:	Shanghai	31.25°N	10:30

Preprocess: remove $p_4 \rightarrow 5$ valid positions.
Cluster: Shanghai $\{p_1, p_3, p_6\}$, Shenzhen $\{p_2, p_5\}$. No cross-edges (1,200 km in 15 min = impossible).

Connected Components



Decision

$n = 5$, threshold $= \max(5, 0.5) = 5$
Significant clusters $= 2$

Result: FALSE
Multiple vehicles detected.

Complexity Analysis & Edge Cases

Phase	Time	Space
Preprocessing	$O(n \log n)$	$O(n)$
Clustering	$O(n \cdot m)$	$O(n)$
Handover Merge	$O(m^3 \cdot b^2)$	$O(m)$
Decision	$O(m)$	$O(1)$
Total	$O(n \log n + m^3 b^2)$	$O(n)$

n = positions, m = clusters ($m \ll n$), $b = 5$.
Typical case ($m < 10$): effectively $O(n \log n)$.

Edge Case Handling

Case	Result
No positions	UNCERTAIN
Single position	TRUE
All invalid	UNCERTAIN
$\Delta t = 0$	100 m tolerance

Input Validation

Null, NaN, and range checks on coordinates and timestamps before processing.

Thank You

Questions?